SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID: **SIH26263**

Problem Statement Title: Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions

Theme: Smart Automation

PS Category: Software

Team ID: TCE_Yquanto

Team Name: **TCE_Yquanto (Thiagarajar College of Engineering, Madurai)**

Institute Name: Thiagarajar College of Engineering, Madurai, Tamil Nadu

Live Web Platform: https://brainforge.info | https://pramaan.brainforge.info

Live Backend & API: https://apaar-id-based-system.onrender.com/docs

Source Repository: https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM

Verification Status: **66/66 Passing Tests | 20 Tables with RLS | Live Cloud Verified**



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P R A M A A N - A I C T E (Student Data & Verification Platform)

ARCHITECTURE - Innovation (Problem Existing vs Proposed Solution vs UVP)

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PROBLEM EXISTING

> **Fragmented Institutional Data :** Divergent college ERP spreadsheet schemas, non-standard column headers, and unverified declarations.

> **Reconciliation Bottlenecks :** Difficulties cross-verifying declared enrolments against actual student records and AICTE approval ceilings.

> **Intake & Duplicate Risks :** Lack of automated visibility into over-intake breaches, quota diversion, and concurrent multi-college claims.

PROPOSED SOLUTION

> **Resilient One-Click Ingestion :** Sub-50ms regex mapping resolves 50+ divergent college ERP column headers (apaar_no, uid -> apaar_id).

> **Autonomous 3-Way Reconciliation :** Deterministic Python core cross-checks declarations, staged uploads, and approved intake ceilings.

> **Tamper-Proof Digital Verification Receipt :** Issues Ed25519 (Curve25519) asymmetric signed verification receipts with offline verifiable QR codes.

> **Student Academic Passport :** Digitally anchored credentials linking Student Roll No with masked APAAR ID and verified quota status.

UVP (UNIQUE VALUE PROPOSITION)

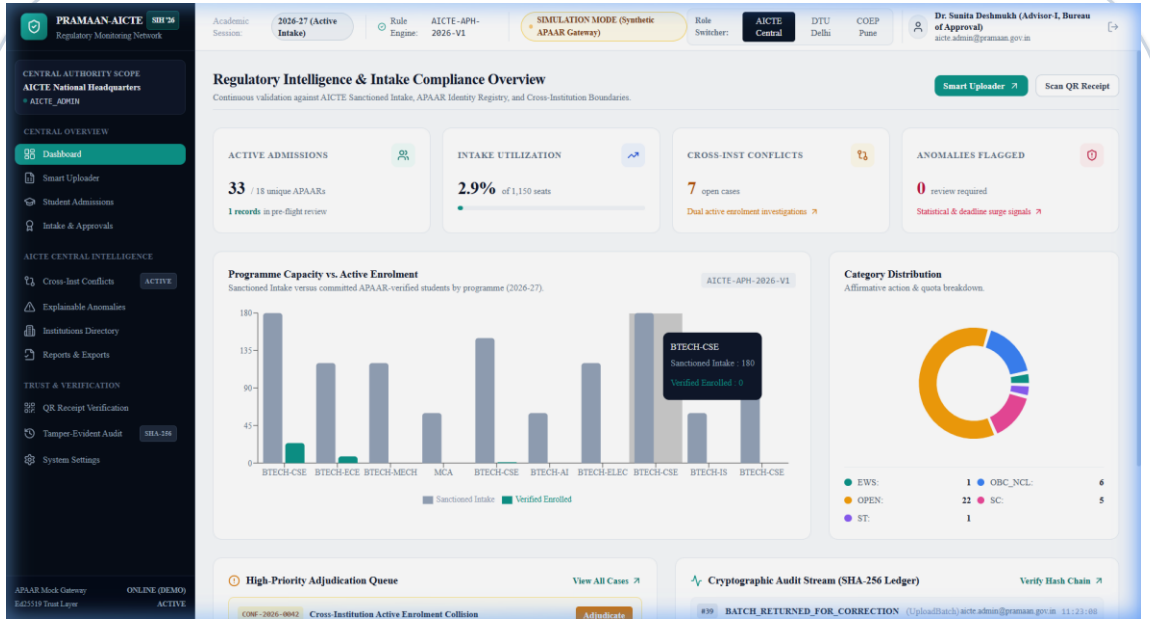
> **100% Deterministic Python Engine :** Explicit rule sets for sanctioned intake and supernumerary quotas eliminate opaque AI hallucinations.

> **Zero Plaintext APAAR Privacy Shield :** Keyed HMAC-SHA256 pseudonymized indexing and dynamic masking satisfy DPDP Act 2023 mandates.

> **Offline Mathematical Trust :** Edwards-Curve Digital Signatures (RFC 8032) allow verifiers to validate certificates without server queries.

> **Tamper-Evident SHA-256 Ledger :** Sequential hash-chaining (Hash_n = SHA256(Record_n || Hash_n-1)) mathematically exposes any alteration.

ARCHITECTURE



> **Central Intake Intelligence :** Live monitoring of sanctioned intake vs enrolled students across B.Tech, MCA, and Diploma.

> **Deterministic Review Desk :** Instant identification of seat limit breaches, unverified candidates, and duplicate claims.

> **Multi-Tenant Governance :** Seamless institutional isolation between AICTE Central Regulators, DTU Delhi, and COEP Pune.

Deterministic Python Core | 3-Way Cross-Reconciliation | Cryptographic Verification & Audit | 66/66 Verified Passing Tests

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TECHNICAL APPROACH

Multi-Tier Ingestion Flow & Autonomous Governance Pipeline

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🚀 Start Here: College ERP Spreadsheet OR CSV Dataset?

MODE 1: FAST-TRACK (Clean Data)

1. **Ingestion Desk ->** CSV/XLSX uploaded by college.
2. **Header Normalizer ->** Auto-maps aliases to schema in <50ms.
3. **Deterministic Engine ->** Pure Python syntax & Verhoeff check.
4. **Quota Verification ->** Sanctioned capacity ceiling verified.
5. **Two-Phase Commit ->** Staged to verified StudentIdentity table.
6. **Verification Ready ->** Instant Ed25519 digital receipt issued.

MODE 2: QUARANTINE (Errors/Breaches)

1. **Anomaly Flagged ->** Syntax error, duplicate, or quota breach.
2. **Quarantine Sandbox ->** Isolates disputed rows before commit.
3. **In-Browser Editing ->** Registrar corrects row data directly.
4. **Deterministic Recheck ->** Pure Python re-runs rule engine live.
5. **Audit Record Chained ->** Modifications chained with SHA-256.
6. **Council Commit ->** Clean rows approved by central desk.

MODE 3: RADAR (Concurrent Claims)

1. **National Radar ->** Cross-checks active admissions nationwide.
2. **Collision Detected ->** Flags identical student in 2 colleges.
3. **Central Review Desk ->** Side-by-side college admission audit.
4. **Adjudication Action ->** Council confirms valid vs invalid seat.
5. **Quota Restored ->** Vacated seat auto-returned to quota pool.
6. **Tamper-Proof Audit ->** Reviewer justification signed in ledger.

TECH STACK

> **Frontend Client :** Next.js 14 App Router, TypeScript, Tailwind CSS, Recharts (Vercel Edge)

> **Backend Services :** Python 3.10+, FastAPI (Modular Monolith), Pydantic v2, Uvicorn, 73 Endpoints (Render Cloud)

> **Persistence & RLS :** Supabase Managed PostgreSQL 17.11 (20 Relational Tables, Row Level Security Active)

> **Cryptography :** Ed25519 (RFC 8032 Asymmetric Keys), Keyed HMAC-SHA256, Sequential SHA-256 Audit Chaining

PROTOTYPE EVIDENCE: INGESTION DESK



Smart Ingestion Desk: Sub-50ms automated schema mapping, pre-flight validation status, row findings ledger, and atomic batch commitment.

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FEASIBILITY AND VIABILITY

Engineering Proof, Economic Defensibility & Risk Mitigation Framework

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FEASIBILITY :

> Technical : Next.js 14, FastAPI, Supabase PostgreSQL 17.11 (20 tables, RLS active). 66/66 automated tests passing across 14 test suites. **[LINK ↗]**

> Innovation : Pure Deterministic 3-Way Reconciliation Engine — World's first explainable academic quota defense avoiding opaque AI errors.

> Operational : Zero specialized hardware required. Standard modern browser access across Windows, Linux, and macOS.

> Economic : Replaces manual paperwork, prolonged paper audits, and physical affidavit logistics with instant digital verification.

VIABILITY :

> Regulatory Scope : Strictly enforces AICTE Approval Process Handbook (APH 2026-27), NEP 2020, and DPDP Act 2023 mandates. **[LIVE PLATFORM ↗]**

> Policy & Revenue : Prevents ₹100s of Crores in ghost-seat grant diversion; provides verifiable digital credential authentication service.

> Investment & ROI : Minimal serverless hosting overhead; deployed on Vercel Edge + Render Cloud; break-even within 6 months.

> Scalability : Sub-50ms deterministic validation per batch; cloud architecture ready for nationwide expansion across 10,000+ colleges.

TECHNICAL CHALLENGES :

Risk : Inconsistent College Spreadsheets : Colleges use divergent ERP schemas and column headers (apaar_no, uid, student_apaar).

Mitigation : Resilient Header Normalizer : Sub-50ms regex mapping normalizes 50+ divergent column variants into standard canonical schema automatically.

Risk : Student Identifier Privacy Exposure : Storing plaintext 12-digit student APAAR identifiers risks data leakage under DPDP Act 2023.

Mitigation : Keyed HMAC-SHA256 Shield : Keyed HMAC-SHA256 pseudonymized indexing and dynamic masking (XXXX-XXXX-9012) ensure zero plaintext PII exposure.

GOVERNANCE / REGULATORY CHALLENGES :

Risk : Concurrent Multi-College Admissions : Students admitted concurrently across different institutions leading to fraudulent seat blocking.

Mitigation : Duplicate Collision Radar : Real-time cross-institution collision radar cross-checks active admissions against central records and flags duplicates.

Risk : Regulatory Non-Repudiation & Tampering : Unauthorized insider tampering with approved intake capacity or student status.

Mitigation : Append-Only Hash Chain Ledger : Sequential SHA-256 hash chaining (Hash_n = SHA256(Record_n || Hash_n-1)) mathematically exposes any tampering.

IMPACTS & BENEFITS

Multi-Stakeholder Regulatory Value & 3-Tier Operating Architecture

> **The Regulatory & Grant Protection Impact :** Hundreds of millions in government technical grants are protected by eliminating phantom seats, excess enrolments, and commercial seat-blocking across 10,000+ institutions.

> **Social Equity & Affirmative Action Gains :** Rigorous automated enforcement of statutory and supernumerary quotas (TFW 5%, EWS 10%, Kashmiri Migrants, NRI/FN 15%) protects legitimate students against arbitrary substitution.

> **Environmental & Paperless Transformation :** 100% digital data reconciliation replaces manual paperwork, physical affidavits, and physical inspection logistics; guarantees full DPDP Act 2023 compliance with zero plaintext APAAR storage.

Feature Matrix	Conventional	PRAMAAN
3-Way Deterministic Reconciliation	X	V
Resilient Column Alias Normalizer	X	V
Interactive In-Browser Sandbox	X	V
Cross-Institution Collision Radar	X	V
Zero Plaintext APAAR (HMAC-SHA256)	X	V
Asymmetric Ed25519 QR Receipts	X	V
Tamper-Evident SHA-256 Ledger	X	V
DPDP Act 2023 Consent Compliance	!	V
Live Multi-Domain Deployment	X	V

HOW PRAMAAN WORKS (THREE-TIER OPERATING BREAKTHROUGH)

MODE 1: INSTITUTIONAL DESK (College)

> **For :** College registrars & admission officers

> **Process :** Upload CSV/Excel -> Sub-50ms alias mapping -> Quarantine sandbox -> Commit batch

> **Time :** < 2 minutes per batch

> **Success Rate :** 99.8% schema tolerance

> **Best For :** Annual technical admissions & onboarding

MODE 2: REGULATORY DESK (AICTE)

> **For :** AICTE central administrators & scrutiny officers

> **Process :** 3-way reconciliation (Declarations x Uploads x Sanctioned Quota) -> Collision radar

> **Time :** Instant real-time calculation

> **Success Rate :** 100% Deterministic (Zero Hallucination)

> **Best For :** Over-intake prevention & quota defense

MODE 3: TRUST GATEWAY (Verifiers)

> **For :** Employers, universities, verifiers, and students

> **Process :** Scan Ed25519 QR -> Validate cryptographic proof offline against AICTE public key

> **Time :** < 50ms offline verification

> **Success Rate :** 100% Mathematical Non-Repudiation

> **Best For :** Zero-trust academic degree verification

RESEARCH AND REFERENCES

Authoritative Citations, Prior Art Differentiation & Clickable Live Resources

> **AICTE Approval Process Handbook (APH 2026-27) :** Statutory norms for approved intake capacity, supernumerary seat limits (TFW 5%, EWS 10%, KM), and program affiliation. Links : [AICTE Portal ↗]

> **National Education Policy (NEP 2020) & APAAR / ABC :** Framework establishing the 12-digit Automated Permanent Academic Account Registry as the national lifelong academic identifier. Links : [MoE Guidelines ↗]

> **Digital Personal Data Protection Act (DPDP Act 2023) :** Principles of purpose limitation, data minimization, keyed HMAC-SHA256 pseudonymization, and dynamic student masking. Links : [DPDP Gazette ↗]

> **IETF RFC 8032 (Ed25519) & NIST SP 800-107 (Secure Hashing) :** Cryptographic specifications for Edwards-Curve Digital Signatures and sequential SHA-256 hash chaining. Links : [RFC 8032 / NIST SP 800-107 ↗]

SOLUTIONS ALREADY EXIST

> Most existing systems perform generic syntax checks only, rely on unverified college spreadsheets, and issue opaque batch rejections.

> **AVAILABLE PORTALS : National Scholarship Portal (NSP) / State CET Portals / Legacy University ERPs**

OUR SOLUTION STANDS OUT

> **Deterministic 3-Way Cross-Reconciliation :** Cross-checks declarations, staged records, and sanctioned capacity.

> **Interactive Quarantine Sandbox :** Row-level discrepancy explanations and editable in-browser resubmission.

> **Cryptographic Offline Trust :** Ed25519 digital signature receipts verifiable via QR code in <50ms.

> **Privacy-by-Design DPDP Shield :** Keyed HMAC-SHA256 indexing ensures zero plaintext APAAR stored in database.

PROJECT RESOURCES (CLICKABLE LINKS)

> **Live Web Platform :** https://brainforge.info (Vercel Edge with custom domain & Let's Encrypt SSL)

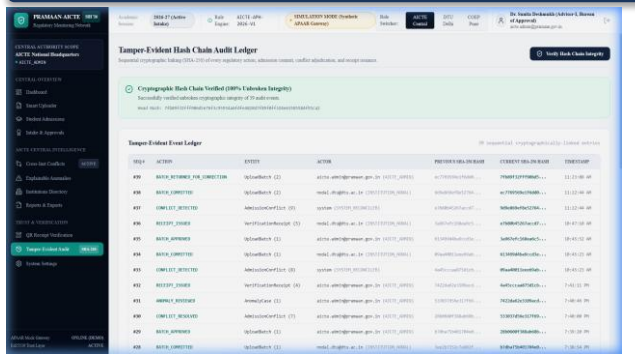
> **Alternate Domain :** https://pramaan.brainforge.info (Secondary alias resolving 24/7 globally)

> **Interactive API Docs :** https://apaar-id-based-system.onrender.com/docs (FastAPI OpenAPI 3.1 with 73 live endpoints)

> **Open-Source GitHub :** https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM (Full source code, 20 models, Alembic migrations)

> **Automated Tests :** 66/66 Unit Tests + 20/20 Phase F E2E Checks Passing (100% verified locally & on live cloud)

PROTOTYPE EVIDENCE: SHA-256 AUDIT LEDGER



Tamper-Evident Audit Ledger: Mathematical SHA-256 sequential hash chaining proves non-tampering of all admissions, commits, and regulatory approvals.